

Cross-Disease Graph Diagnosis: A 26-Dataset Benchmark of CNN and GNN Architectures and a Falsifiable Test of Cross-Disease Patient Bridges

MEGA-27 Item 10 — Deep-Learning Diagnosis Suite

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Abstract

We benchmark MLP, 1D-CNN, GCN and GAT diagnosis models against strong classical baselines on 26 real clinical datasets (accession-level, all URLs verified), with bootstrap confidence intervals and calibration metrics, and we test a falsifiable discovery claim: cross-disease *patient bridges* — edges of a unified patient-similarity graph spanning diseases built on disease-agnostic distributional fingerprints — carry transferable diagnostic signal. All claims are reported with seed variance; negative results are preserved.

1 Introduction

2 Methods

2.1 Problem setting

Each dataset d is a cohort $(X^{(d)}, y^{(d)})$ with n_d patients and p_d native features; $y_i^{(d)} \in \{0, 1\}$ is the diagnosis.

2.2 Models

MLP, 1D-CNN over the feature axis, GCN and GAT over a patient k NN graph.

2.3 Unified cross-disease graph

Per-disease encoders $E_d : \mathbb{R}^{p_d} \rightarrow \mathbb{R}^{16}$ map patients into a shared latent space; the node input is $[E_d(x) \parallel \phi(x)]$ where ϕ is the 10-dimensional distributional fingerprint (Eq. 3).

2.4 Mathematical formulation

Gaussian k NN kernel with self-tuned bandwidth:

$$A_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right), \quad \sigma^2 = \text{median}_{i,j \in \text{kNN}(i)} \|x_i - x_j\|^2. \quad (1)$$

Symmetric GCN normalization (Kipf & Welling):

$$\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}, \quad \tilde{D}_{ii} = 1 + \sum_j A_{ij}. \quad (2)$$

GCN layer: $H^{(l+1)} = \tanh(\hat{A}H^{(l)}W^{(l)})$. Distributional fingerprint (disease-agnostic; comparable across diseases):

$$\phi(x) = (\mu, \sigma, \min, \max, q_{25}, q_{50}, q_{75}, m_3, m_4, \|z\|_1/d), \quad m_r = \frac{1}{p} \sum_j z_j^r, \quad z_j = (x_j - \mu)/\sigma. \quad (3)$$

Class-weighted BCE with $w_1 = n_-/n_+$:

$$\mathcal{L} = -\frac{1}{n} \sum_i w_{y_i} [y_i \log \sigma(f_i) + (1 - y_i) \log(1 - \sigma(f_i))]. \quad (4)$$

Multi-task transductive objective over diseases $\mathcal{D}$ and train masks T_d :

$$\mathcal{L}_{\text{MT}} = \sum_{d \in \mathcal{D}} \mathcal{L}^{(d)}(T_d), \quad \nabla_{\theta} \mathcal{L}_{\text{MT}} = \sum_d \nabla_{\theta} \mathcal{L}^{(d)}. \quad (5)$$

Expected calibration error:

$$\text{ECE} = \sum_{b=1}^B \frac{|S_b|}{n} |\text{acc}(S_b) - \text{conf}(S_b)|. \quad (6)$$

Percentile bootstrap CI: for $B = 1000$ resamples,

$$\text{CI}_{95} = [Q_{0.025}(\theta_b^*), Q_{0.975}(\theta_b^*)]. \quad (7)$$

Bridge ablation effect (paired per seed s):

$$\Delta^{(d)} = \frac{1}{S} \sum_s (\text{AUC}_{s,\text{full}}^{(d)} - \text{AUC}_{s,\text{no-bridge}}^{(d)}). \quad (8)$$

Mann–Whitney form of the AUC:

$$\text{AUC} = P(f(x^+) > f(x^-)) + \frac{1}{2}P(f(x^+) = f(x^-)). \quad (9)$$

Bridge homophily (label agreement across a bridge):

$$h = \frac{|\{(i, j) \in E_{\text{cross}} : y_i^{(d_i)} = y_j^{(d_j)}\}|}{|E_{\text{cross}}|}. \quad (10)$$

Assumptions for a sign test. An exact binomial sign test is valid only when the nonzero paired differences are independent and their signs have probability one-half under the null. Repeated seeds sharing a cohort, overlapping train sets and graph construction do not automatically satisfy independence or symmetry. Neither a no-signal hypothesis nor seed randomness proves these assumptions. A bridge-ablation contrast is descriptive until its dependence structure and null construction are justified; the previously asserted unconditional falsifier is withdrawn.

3 Datasets

4 External tools

5 Results

6 Discovery experiment: cross-disease bridges

7 Negative results

7.1 Locked pneumonia Model C: three-seed negative

The progressive CXR transfer experiment completed the three fixed seeds. Test accuracies for seeds 11, 23 and 37 were respectively 0.8605769231, 0.9070512821 and 0.9198717949. Mean accuracy was 0.8958333333 with sample standard deviation 0.0311985935. The two-sided 95% Student-*t* interval across three seeds was [0.8183317307, 0.9733349360]. The mean did not reach the Model C hypothesis threshold 0.90, and neither the mean-above-0.928 condition nor the interval-excluding-0.928 condition held. There is no benchmark beat.

Seed 37 used a validation-only threshold of 0.59, fixed before recovery test inference. Its test AUC was 0.9628205128 and Brier score 0.0679723889. An original evaluation process was killed at its 120-second execution limit without recovered output; one recovery sequence retained the identical frozen checkpoint and evaluation rule, persisted batch predictions and left seeds 11 and 23 unchanged. This technical interruption is disclosed separately from the measured result. AUC does not replace the accuracy gate.

The original evaluator omitted preregistered ECE; the three-seed reporting gap remains explicit. A post-hoc description of already evaluated seed37 predictions gives ten-fixed-bin ECE 0.0468606536. At its frozen threshold the confusion counts are TP377, TN197, FP37 and FN13 on 624 images; sensitivity is 0.9666666667, specificity 0.8418803419 and alert rate 0.6634615385. These figures are descriptive, not new inference, a three-seed calibration result or a preregistered ECE test. Seed11/23 prediction archives were not found. Model D’s census-cleaning ablation is separate unfinished work. This three-seed result closes Model C, not all diagnosis-suite requirements, and does not justify changing a locked threshold or tuning on test data.

7.2 Malaria R-M2: stronger backbone does not pass

A ResNet18 trained from scratch (weights=None), under the fixed matched R-M1 protocol, reached mean fold accuracy 0.9471001905, SD 0.0128463226, and 95% *t* interval [0.9311493743, 0.9630510067]. The gate requiring mean accuracy strictly greater than 0.957 and an interval excluding that comparator failed. Its paired mean change from R-M1 was -0.0005998095 ; the recorded paired bootstrap interval $[-0.0103682989, 0.0069915942]$ includes zero. This is not a pretrained-backbone experiment and not evidence that backbone size is the cause of failure. Source: subitems_4.5/results/malaria/rm2/VERDICT.json.

7.3 Completion boundary

This paper remains a short working draft. The dataset and tool census, full comparator verification, bridge-discovery evidence and the 50-page final artifact gates are open. Empty section headings do not constitute measured results or counted datasets. The negative pneumonia and malaria arms are closed without threshold changes; they do not establish a suite-level benchmark win.

8 Conclusion

References

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